

Online Review Sentiment Analysis of Quickbite Food Delivery App

1. Introduction

The following report is a sentiment analysis of customer reviews of the mobile food delivery application, "QuickBite." The challenge is to create an NLP pipeline to automatically categorize raw text reviews into 3 sentiment categories: Positive, Negative and Neutral. Simulated and synthetic reviews were used, generating 150 reviews to mimic real user feedback from the App Store and Twitter. The analysis can help to gauge customer satisfaction and provide a look into areas for service improvement.

2. Methodology

2.1 Dataset: 150 reviews were created with the help of AI templates, including 60 positive, 51 negative, and 39 neutral sentiments. Eight product categories had reviews collected from the App Store (85) and Twitter (65).

2.2 Preprocessing: Text data was preprocessed to lower the case of text and to extract the URL, mention, hashtag, punctuation, and stop words (such as words the, is, and like). The tokenization was carried out via NLTK.

2.3 Feature extraction: TF-IDF (Term Frequency-Inverse Document Frequency) was applied, which has max_features=3000 and ngram_range= (1,2). Single words (unigrams) and phrases of two words (bigrams) such as "customer support" or "cold pizza" are captured.

2.4 Model Training: We trained three classification models: logistic regression, multinomial naive Bayes, and linear SVM. Stratified sampling was employed to ensure the proportion of classes in the data was 80:20.

3. Model Comparison and Evaluation

The models were evaluated on the same 30% test set. The results are summarized below:

Model	Accuracy	Precision	Recall	F1-Score
Linear SVM	96.67%	96.97%	96.67%	96.63%
Logistic Regression	93.33%	94.28%	93.33%	93.11%
Multinomial Naïve Bayes	93.33%	93.89%	93.33%	93.00%

Recommendation: Use linear SVM model. All models worked well as the dataset was synthetic and clean, but SVM had the maximum F1 score. Here we choose to use F1-score since it considers both precision and recall, and we don't want the model to only focus on the majority class (positive reviews).

4. Confusion Matrix Analysis

The linear SVM model's confusion matrix shows that: The model perfectly classified all 12 positive reviews and 10 negative reviews. There was only one miss with neutral reviews, 1 out of 8 neutral reviews were misclassified as negative. This is a typical problem in sentiment analysis. Neutral reviews may include ambiguous language (e.g., the food was okay, delivery was okay), that is common to both extremes and neutrals. This could be remedied by adding more sophisticated neutral examples to the data set.

5. Sentiment Trends

The visualizations are available as attachments in the Jupyter Notebook. The overall distribution is positive reviews (40%), neutral (26%) and negative (34%). This means overall satisfaction, but it also shows a substantial group of dissatisfied users. Trend Over Time: Sentiment changes monthly. The highest sentiment was in March and September. Negative sentiment rose sharply in August, which may be the result of seasonal effects (such as hot weather affecting food temperature, or delayed deliveries). By Source: Twitter has a higher proportion of negative reviews (~37%) compared to the App Store (~30%). That implies that individuals who are on social media platforms are more likely to be frustrated publicly than the ones who leave out formal app reviews. By Product: Dessert and sushi had mainly positive responses. There are a few noteworthy negative reviews for Beverage, possibly because of spills or packaging in transit.

6. Limitations:

Synthetic Data: The dataset is created from templates, which results in a very clean data set. The model's accuracy to real world text is diminished by the presence of slang, typos and complex grammar in it.

Sarcasm Detection: Words such as "Great! Another cold pizza" is negative pizza, but positive words. The TF-IDF model does not have a mechanism to recognize sarcasm; it is simply based on word frequency, not context.

Class Imbalance (Positive > Negative > Neutral): The dataset is slightly imbalanced which can lead to the model predicting more Positive.

7. Future Improvement

Word embeddings (Word2Vec or GloVe) and transformer models such as BERT could be used to enhance the model. BERT does not rely on the TF-IDF approach and is able to recognize sarcasm. Besides, pooling a larger real-world real data set with human annotation would give a more realistic evaluation.

8. Business Implications and Conclusion

This sentiment analysis indicates that most customers are pleased with Quickbite, but there is a consistent problem with delivery, temperature and speed. The company should: Invest in good food delivery packaging, particularly for food and beverage and pizza to keep food hot and fresh. Keep an eye on Twitter—because negative sentiment tends to be more prevalent there, it's worth having a dedicated social media response team who can respond to complaints on a timely basis so that your brand's reputation is not damaged. Review and examine the sentiment dip in August; this could be related to the very hot summer, delivery bags with insulation which may help resolve this.